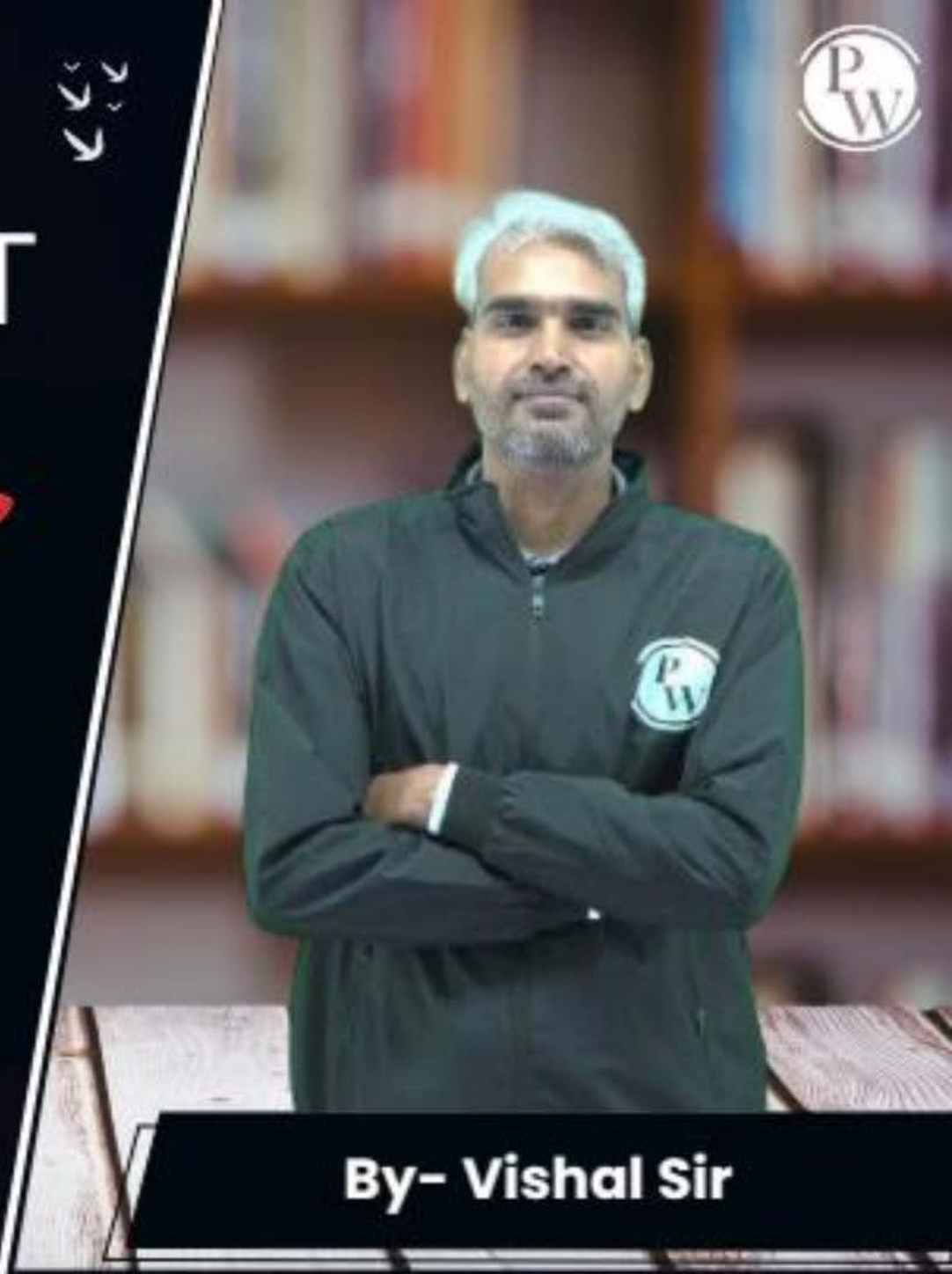
Computer Science & IT



Discrete Mathematics

Graph Theory

Lecture No. 3



By- Vishal Sir

Recap of Previous Lecture



1. Topic — Representation of graph
2. Topic — Simple graph, multi graph and pseudo graph
3. Topic — Maximum number of edges in a simple graph
4. Topic — Maximum number of simple graph with n vertices
5. Topic — Terminologies related to graphs
6. Topic — Different types of graphs

Slide

Topics to be Covered



- Topic: Bi-partite graph
- Topic: Complete bi-partite graph
- Topic: Sum of degree theorem
- Topic: Degree sequence
- Topic: Havel Hakimi's result

Slide

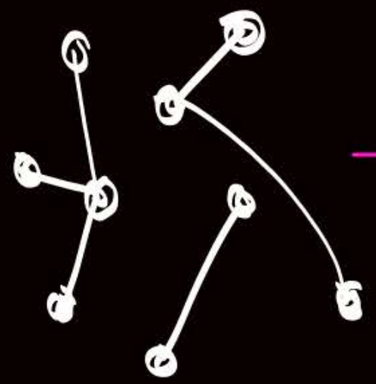


Topic : Types of graphs

1. Complete Graph
2. Cycle Graph
3. Wheel Graph
4. Connected Graph
5. Cyclic Graph
6. Acyclic Graph
7. Tree
8. Bipartite Graph
9. Complete Bipartite Graph

H.W.

Q: A Forest with n vertices and 'K' Components i.e. 'K trees' will have how many edges ?



# Vertices	# trees	# Edges
n	1	$(n-1)$
n	2	$(n-1)-1 = n-2$
n	3	$(n-2)-1 = n-3$
⋮	⋮	⋮
n	k	$(n-k)$ Ans



Topic : Bi-partite Graph

i.e: No two vertices within the same set V_1 or V_2 should be adjacent to each other

- A bipartite graph (or bigraph) is a graph whose vertices can be partitioned into two sets V_1 and V_2 such that every edge of the graph connects a vertex in set V_1 to a vertex in set V_2.

Let $G=(V, E)$, if Partition of $V=\{V_1, V_2\}$, then $V_1 \cap V_2 = \emptyset$, $V_1 \cup V_2 = V$

"Ordered triple"

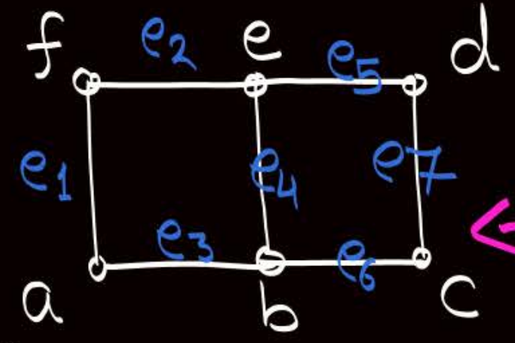
A bipartite graph can be represented using the notation $G=(V_1, V_2, E)$ where,

V_1 is the first set of the partition

V_2 is the second set of the partition and

E is the set of edges of graph G

eg:



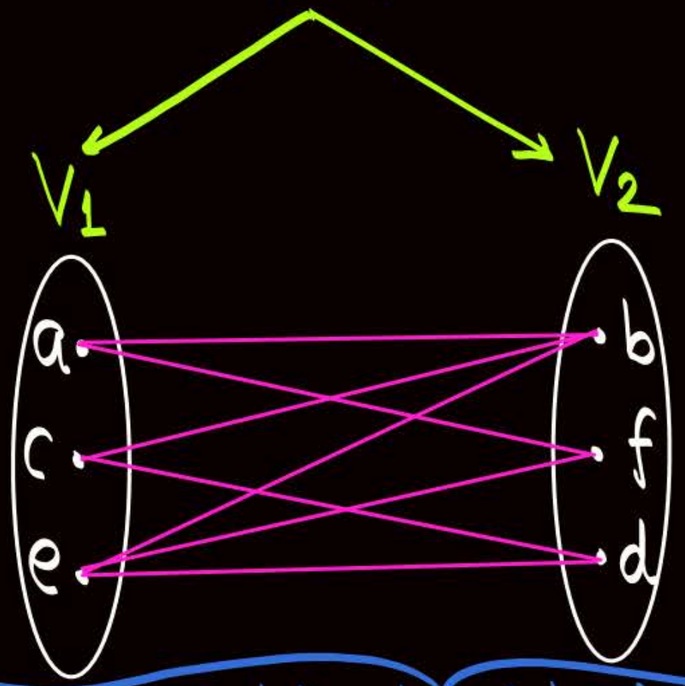
$G = (V, E)$

$V = \{a, b, c, d, e, f\}$

$E = \{e_1, e_2, e_3, e_4, e_5, e_6, e_7\}$

Check whether graph G is a bi-partite graph or not

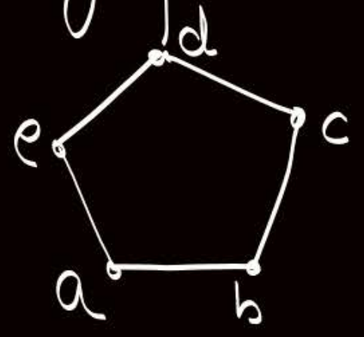
$V = \{a, b, c, d, e, f\}$



We could partition the set of vertices into two subset such that each edge of the graph connect a vertex of first subset to a vertex of another subset. {i.e, No two vertices of same subset are adjacent to each other}

∴ Given graph is a bipartite graph

Q: Check whether the following graph is a bi-partite graph or not.



$V = \{a, b, c, d, e\}$

$V = \{a, b, c, d, e\}$

V_1 V_2

a. c. b. e.

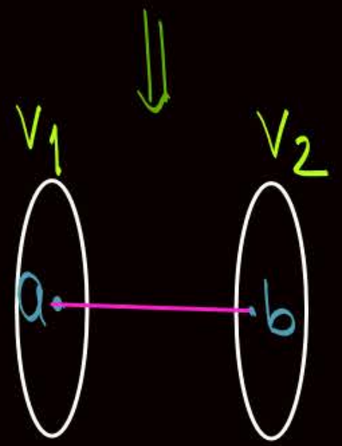
Vertex 'd' can not be placed in subset V_1 because it is adjacent to vertex 'c'; similarly vertex 'd' can not be placed in subset V_2 because it is adjacent to 'e'

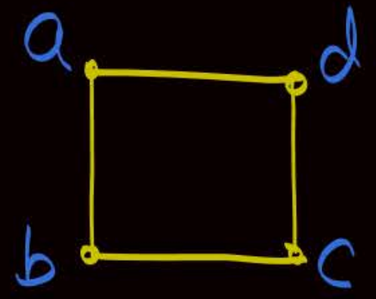
Vertex 'd' can not be placed in any subset

↳ i.e. Vertex set V can not be partitioned into two subset as per given constraint ∴ Graph is not a bipartite graph

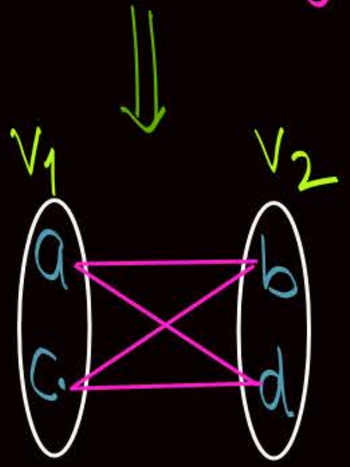


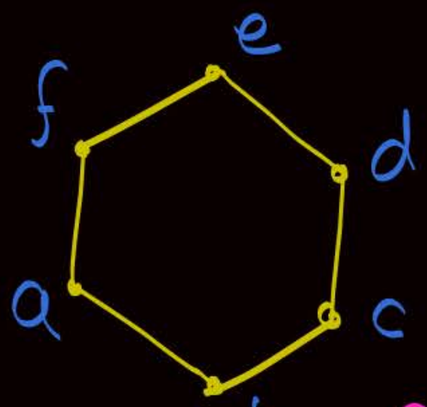
G_1 { cycle length = 0 ↑ Even }



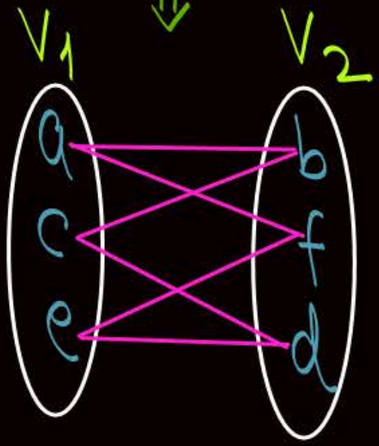


G_2 { cycle length = 4 ↑ Even }





G_3 { cycle length = 6 ↑ Even }



Note:-

① In a graph if all cycles are of even length. then that graph is always a bipartite graph

② Tree is an acyclic graph { i.e. cycles of length = 0 (Even) }
∴ Every tree is a bipartite graph

③ If there exist any cycle of odd length, then graph can not be a bipartite graph

Note:

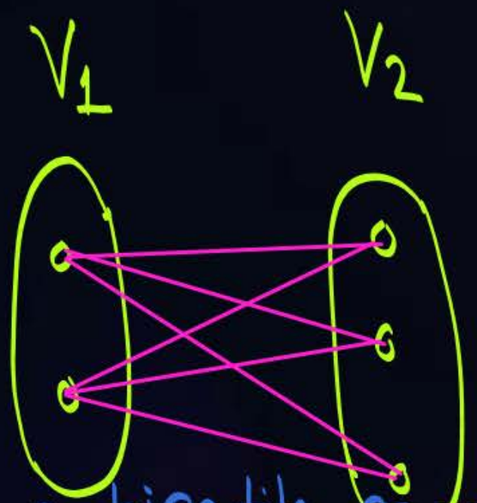
④ A graph with a single vertex can never be a bipartite graph

⑤ Every 'Null graph' with two or more vertices is a bipartite graph



Topic : Complete bipartite Graph

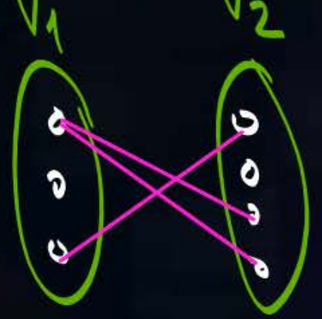
- A bipartite graph $G=(V_1, V_2, E)$ is called a complete bipartite graph if every vertex in set V_1 is adjacent to every vertex of set V_2.



It is a bipartite graph,
And Every vertex of subset V_1 is adjacent to Every vertex of subset V_2
∴ Complete bipartite graph

Every Complete bi-partite graph is also a bipartite graph,
but Every bipartite graph need not be Complete bipartite graph

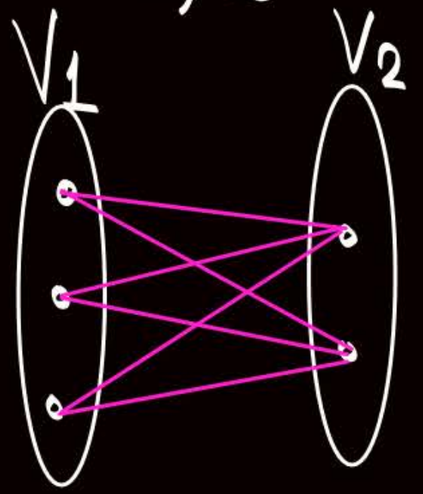
eg



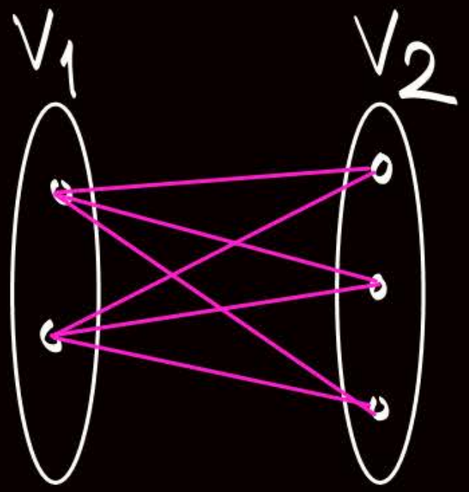
Bipartite ✓
Complete Bipartite ✗

* A complete bipartite graph $G = (V_1, V_2, E)$ where $|V_1| = m$ and $|V_2| = n$ is represented by $K_{m,n}$

$K_{3,2}$



$K_{2,3}$

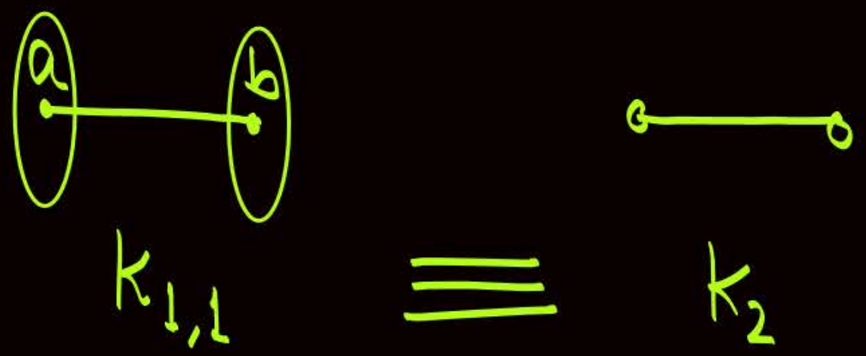


* In a Complete bipartite graph $K_{m,n}$

$$|E(K_{m,n})| = m \cdot n$$

In general, Complete bipartite graph is not a Complete graph

{ A complete bipartite graph $K_{m,n}$ is a Complete graph if and only if '$m=n=1$'



Q:- Let G be a bipartite graph with 'n' vertices, $|V_1|+|V_2|=n$
then maximum number of edges possible in graph G is?

let $|V| = 10$

$\lvert V_1 \rvert$	$\lvert V_2 \rvert$	Max No. of Edges Possible
1	9	9
2	8	16
3	7	21
4	6	24
5 $\left(\frac{10}{2}\right)$	5 $\left(\frac{10}{2}\right)$	25 $\left(\frac{10}{2} * \frac{10}{2}\right)$
6	4	24
7	3	21
8	2	16
9	1	9

let $|V| = 10$

$\lvert V_1 \rvert$	$\lvert V_2 \rvert$	Max No. of Edges Possible
1	8	8
2	7	14
3	6	18
4 $\left(\frac{9-1}{2}\right)$	5 $\left(\frac{9+1}{2}\right)$	20 $\left(\frac{9-1}{2}\right) * \left(\frac{9+1}{2}\right)$
5 $\left(\frac{9+1}{2}\right)$	4 $\left(\frac{9-1}{2}\right)$	20 $\left(\frac{9+1}{2}\right) \cdot \left(\frac{9-1}{2}\right)$
6	3	18
7	2	14
8	1	8

Q:- Let G be a bipartite graph with 'n' vertices, then maximum number of edges possible in graph G is?

Maximum no. of Edges Possible

- $\frac{n^2}{4}$ if n = Even
- $\left(\frac{n+1}{2}\right)\left(\frac{n-1}{2}\right)$ if n = odd

Maximum no. of Edges Possible $= \left\lfloor \frac{n^2}{4} \right\rfloor$

K-partite graph :-

If we can partition the set of vertices into 'k' non-empty subsets such that no two vertices of same subset are adjacent to each other, then it is a k-partite graph

* If graph is a k-partite graph then graph is (k+1) partite as well { Provided (k+1) is less than or equal to the total number of vertices in the graph }

② If graph is a k-partite graph, then it need not be (k-1) partite



Topic : Sum of degree theorem / Handshaking lemma

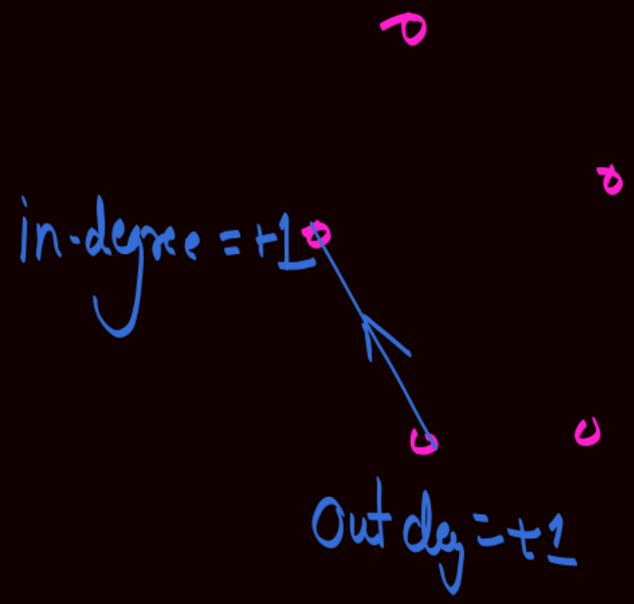
Let $G=(V,E)$ be a graph

where $V=\{v_1, v_2, v_3, \ldots, v_{|V|}\}$

then

$$\sum_{i=1}^{|V|} \deg(v_i) = 2\,|E|$$

$$\deg(v_1) + \deg(v_2) + \deg(v_3) + \cdots + \deg(v_{|V|}) = 2 * |E|$$


in-degree = +1
Out deg = +1

Corollary 1:-

If graph $G = (V, E)$ is a directed graph,

then $$\sum_{i=1}^{|V|} deg^{+}(v_i) = \sum_{i=1}^{|V|} deg^{-}(v_i) = |E|$$

Corollary 2:-

We know $\sum_{i=1}^{|V|} \deg(v_i) = 2|E|$

it is always even

:- In any graph G, total number of vertices of 'odd' degree will always be even

Corollary 3:

In a graph G if degree of Each vertex is exactly 'K' { i.e. graph is K-regular }

then $$K \cdot |V| = 2|E|$$

Logic:

$$\deg(v_1) + \deg(v_2) + \cdots + \deg(v_{|V|}) = 2|E|$$

$$\underbrace{K + K + \cdots + K}_{|V| \text{ times } K} = 2|E|$$

$$\therefore K|V| = 2|E|$$

Corollary 4

In a graph G if degree of Each vertex is at least 'K' {i.e. greater than or equal to K}

└ i.e. $\deg = K$ or (K + some non-negative integer)

then $\boxed{K|V| \leq 2|E|}$

Logic:- $\deg(v_1) + \deg(v_2) + \cdots + \deg(v_{|V|}) = 2|E|$

$(K+x_1) + (K+x_2) + \cdots + (K+x_{|V|}) = 2|E|$ Where $x_i \geq 0, \forall i$

Where $x_1 \geq 0$ ↑ integer

$$K \cdot |V| + \sum_{i=1}^{|V|} x_i = 2|E|$$

└ $\forall i, x_i \geq 0$, $\therefore \sum_{i=0}^{|V|} x_i \geq 0$

Corollary 5 :-

In a graph G if degree of Each vertex is at most K.

then $$K|V| \geq 2|E|$$

$$\deg(v_1) + \deg(v_2) + \cdots + \deg(v_{|V|}) = 2|E|$$

$$(K - x_1) + (K - x_2) + \cdots + (K - x_{|V|}) = 2|E| \quad \text{when } x_i \geq 0, \forall i$$

$$K|V| - \left\{ \sum_{i=0}^{|V|} x_i \right\} = 2|E|$$

$$\left\{ \sum_{i=0}^{|V|} x_i \right\} \geq 0$$

Note:

In a graph G.

degree is at least $\delta(G)$ $\therefore$ $\delta(G) \cdot |V| \leq 2|E|$ — ①

and degree is at most $\Delta(G)$ $\therefore$ $\Delta(G) \cdot |V| \geq 2|E|$ — ②

By ① & ②

$$\delta(G) \cdot |V| \leq 2|E| \leq \Delta(G) \cdot |V|$$

H.W

#Q. The number of edges in a k - regular graph with n vertices is

H.W.

#Q. A non-directed graph contains 16 edges and all vertices are of degree 2. Then the number of vertices in G is _____.

#Q. G is undirected graph with n vertices and 25 edges such that each vertex of G has degree at least 3. Then the maximum possible value of n is _____.

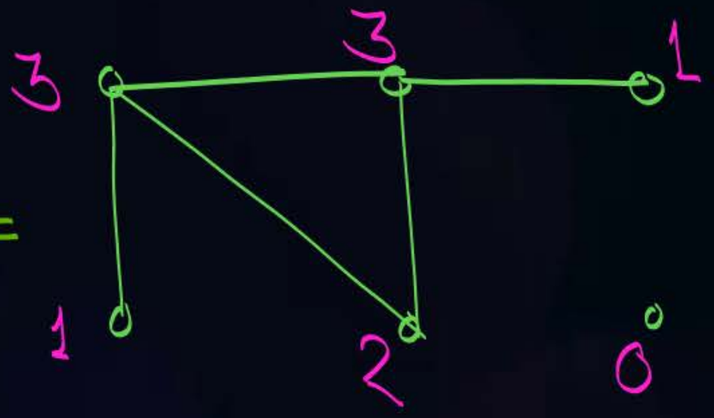


Topic : Degree sequence

In a graph G if degrees of all the vertices are arranged in non-increasing or non-decreasing order, then it is called degree sequence of graph G.

eg:

Graph G =



Degree Sequence of graph G in non-increasing order = (3, 3, 2, 1, 1, 0)

Degree Sequence of graph G in non-decreasing order = (0, 1, 1, 2, 3, 3)

Graphic:- If there exists a simple non-directed graph w.r.t given degree sequence then that degree sequence is called a graphic, Converse of the statement is also true

#Q. Which of the following degree sequences represent a simple non-directed graph? {i.e. which of the following is/are graphic?}

1. {2, 3, 3, 4, 4, 5} ⇒ No. of vertices with odd degree can never be odd in any graph (simple + not-simple).
2. {2, 3, 4, 4, 5}
3. {1, 3, 3, 3}
4. {0, 1, 2, 3,.....,n-1}
5. {1, 3, 3, 4, 5, 6, 6}
6. {3, 3, 3, 3, 2}

∴ Given degree sequence does not represent any graph. Hence it is not "a graphic".

#Q. Which of the following degree sequences represent a simple non-directed graph?

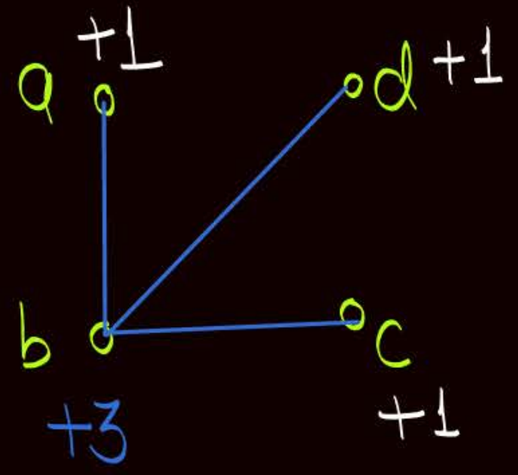
1. {2, 3, 3, 4, 4, 5}
2. {2, 3, 4, 4, 5}
3. {1, 3, 3, 3}
4. {0, 1, 2, 3,.....,n-1}
5. {1, 3, 3, 4, 5, 6, 6}
6. {3, 3, 3, 3, 2}

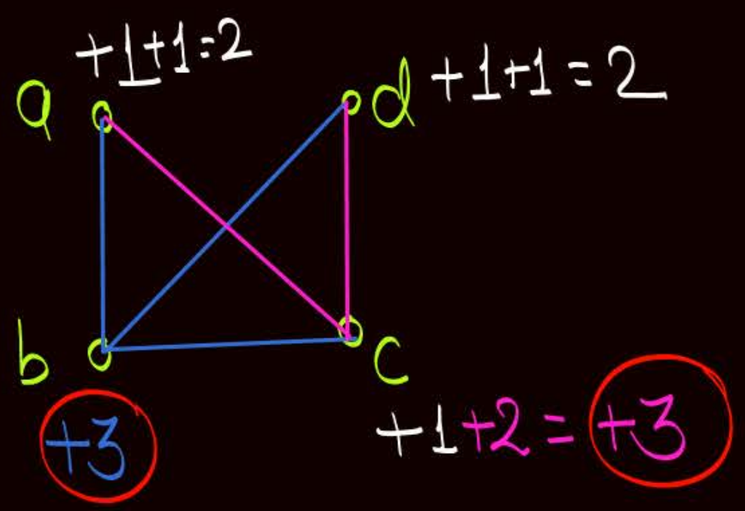
In a simple graph with n-vertices, degree of every vertex $\leq (n-1)$

There are '5' degrees in the sequence i.e. '5' vertices in the graph

└ In a simple graph with '5' vertices degree of a vertex can be at most '4' $(5-1)$

└ ∴ Vertex of degree = 5 is not possible





no. of vertices "n=4"

∴ Max degree possible for a vertex in a simple graph = (4-1) = 3

Note:- In a simple graph with 'n' vertices, if there are 'k' vertices of degree = (n-1) then in that simple graph degree of every vertex must be 'at least k'

#Q. Which of the following degree sequences represent a simple non-directed graph?

1. {2, 3, 3, 4, 4, 5}
2. {2, 3, 4, 4, 5}
3. {1, 3, 3, 3}
4. {0, 1, 2, 3,.....,n-1}
5. {1, 3, 3, 4, 5, 6, 6}
6. {3, 3, 3, 3, 2}

No. of vertices = 4

And there are '3' vertices of deg = 3 {i.e 4-1} (n-1)

↳ Hence in a simple graph degree of Every Vertex must be at least '3'

↳ And hence Vertex of deg = 1 Can not Exist

#Q. Which of the following degree sequences represent a simple non-directed graph?

1. {2, 3, 3, 4, 4, 5}
2. {2, 3, 4, 4, 5}
3. {1, 3, 3, 3}
4. {0, 1, 2, 3,.....,n-1}
5. {1, 3, 3, 4, 5, 6, 6}
6. {3, 3, 3, 3, 2}

Total no. of vertices = 'n'
And there is '1' vertex of degree = (n-1)

i.e. In a simple graph degree of every vertex must be at least '1'
Hence vertex of deg = 0 can not exist

Not true for a graph which is not a simple

Eg: deg:1 deg=3

Note: In a simple graph all vertices can not have distinct degrees.
i.e.; In a simple graph there must be at least two vertices of same degree

#Q. Which of the following degree sequences represent a simple non-directed graph?

1. {2, 3, 3, 4, 4, 5}
2. {2, 3, 4, 4, 5}
3. {1, 3, 3, 3}
4. {0, 1, 2, 3,.....,n-1}
5. {1, 3, 3, 4, 5, 6, 6}
6. {3, 3, 3, 3, 2}

No. of vertices = 7
and we have '2' vertices of deg = 6 = (7-1) = (n-1)

∴ In a simple graph degree of Every vertex must be at least '2'.

Hence in a graphic vertex of deg = 1 is not possible

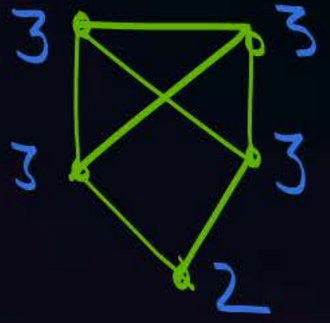
#Q. Which of the following degree sequences represent a simple non-directed graph?

1. {2, 3, 3, 4, 4, 5}
2. {2, 3, 4, 4, 5}
3. {1, 3, 3, 3}
4. {0, 1, 2, 3,.....,n-1}
5. {1, 3, 3, 4, 5, 6, 6}
6. {3, 3, 3, 3, 2}

None of Conditions based on which we can say that "it is not a graphic" is satisfied.

∴ We will have to check Explicitly.

We can observe that there is a Simple graph w.r.t. given degree Sequence.



Hence given degree sequence is a graphic!

#Q. Which of the following degree sequences represent a simple non-directed graph? (OR) Which of the following deg. sequences is/are graphic?

1. {2, 3, 3, 4, 4, 5}
2. {2, 3, 4, 4, 5}
3. {1, 3, 3, 3}
4. {0, 1, 2, 3,……,n-1}
5. {1, 3, 3, 4, 5, 6, 6}
6. {3, 3, 3, 3, 2}



2 mins Summary

Topic Bi-partite graph

Topic Complete bi-partite graph

Topic Sum of degree theorem

Topic Degree sequence

Topic Havel Hakimi's result

Next Class

Slide



THANK - YOU

Slide